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ICT COMPETENCY FRAMEWORK FOR SECONDARY SCHOOLS

SECONDARY 1 TO 3

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The National ICT Curriculum team members comprised of the following:

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***Note**: Titles and locations of all persons listed in the Acknowledgments were current when this document was being prepared.*

FOREWORD

For years now, Information and Communication Technology (ICT) has found its way into our educational system, and it is here to stay. ICT competencies are seen by many countries to be the vital core of their educational system along side the "3 Rs" of education which are Reading, Writing and Arithmetic. One of the overriding aims of the Ministry of Education is to prepare our children educationally so that they can play full roles in a modern society.

This curriculum framework, developed in 2005/2006 by a team of committed ICT education teachers has two key purposes. The first purpose is to assist the Seychelles government to implement its policy towards technology education and should be seen as the government's continuous effort to integrating the new technologies into its educational system. Secondly, it specifies a curriculum in ICT for our secondary schools, that is in line with current international trends and standards. The framework establishes ICT competencies expected of all students from Secondary 1 (S1) to Secondary 3 (S3). It aims to guarantee that all secondary school students in the Seychelles develop their best in the area of ICT education, which would ultimately enable them to transfer their skills across the curricula and to the society at large.
The framework document should not be seen as a document which orders and stipulates what students should learn. Rather it should be used by the school as a resource and guide for schools when developing the teaching and learning programmes to meet the needs of their students.

The Audience For This Framework

This framework is developed for several overlapping audiences. One audience comprises secondary school ICT education teachers and the planners and developers of ICT education programmes and resources, all at many different levels. Another one is those primary school teachers interested in ICT education. Perhaps most importantly, are all readers who are concerned about ICT education and development in our education system.
The frameworks should be regarded as a 'dictionary' of competencies, rather that as a description of required competencies for all levels.

FRAMEWORK SUMMARY

This curriculum document is the first of its kind in terms of approach, in the history of curriculum development for ICT in the Seychelles. Conducted under the auspices of the National Institute of Education, the Curriculum Team was formed by a group of teachers selected for their great interest, deep commitment, and years of experience in using computers in a variety of settings in education. The ultimate aim of the curriculum team was to produce a framework document for ICT that could serve as a hands-on teaching guide, reference and resource for teachers

There is now a world-wide movement towards competency-based learning, and Seychelles is undergoing dramatic transformations in response to this movement. For the purposes of this framework document, a competency is defined as a "combination of skills, knowledge, and attitudes needed to perform a specific task." This Competency based curriculum, then, contains policy statements and guidelines directed at defining, teaching, and assessing ICT competencies across our secondary schools.

More specifically, the broad objectives of the framework are as follows:

- Promote the harmonization of activities, approaches and standards in the educational uses of Information and Communications Technology within our schools.
- Foster the concept of Life Long Learning among students and teachers through the use of the World Wide Web and Email.
- Ensure that there exists equitable access to ICT resources by students within or schools.
- Demonstrate the Ministry Of Education's intention at providing a sound level of Computer Literacy to students.

Why is it imperative to implement competency-based initiatives for ICT curriculum?

- Firstly, specific competencies help schools, teachers and students, as well as other stakeholders such as employers and Further Educational Institution, to have a common understanding about the specific skills and knowledge about ICT that student should master as a result of their learning experiences in the schools.
- Secondly specific articulations of competencies inform and guide the basis of subsequent assessments at the school.
- Last but not least, specific competencies provide directions for designing learning experiences and assignments that will help students gain practice in using and applying these competencies in different contexts.

Rationale and Philosophy

Throughout the world, information and communications technology (ICT) is changing the face of education. Two fundamental and complementary factors are at work. First, ICT is changing the nature of work and the workplace, (Kull, M. and Halal, 1998, p.2) and education systems must respond to this. The so-called "knowledge revolution", combined with economic globalization, create conditions which strongly reward those countries that focus growth on knowledge-based industries. A prerequisite for this is an educated labour force of computer-literate individuals who themselves understand and can harness the power of the ICT revolution. More generally, every citizen in this and the next generation will need to have a high "comfort level" with technology to live in and contribute to a society increasingly part of an interdependent "wired world". So ICT is changing the objectives of education.

Second, ICT provides educators with a powerful new tool to enhance the learning opportunities for students and the professional development opportunities for teachers. Thus ICT is also changing the methodologies through which educational services are delivered.

The Framework is grounded in research, and theoretical literature of the ICT technologies in teaching and learning. The framework focuses on contemporary ways of thinking about the way people learn.

The Ministry of Education and the National ICT Policy state that:

- Special emphasis should be given to ICT literacy in order to equip the countries workforce with the basic fundamental skills required for success in an increasingly technological driven business environment. *(Seychelles Strategy 2017* **P. 27***)*
- In the light of advances in Digital Educational Technology a constant re-examination of teaching methods, strategies and content that will incorporate ICT will have to be constantly made. (*EDUCATION FOR A LEARNING SOCIETY: Policy statement of the ministry of education, Seychelles* **P. 2**)
- A mastery of modern media of communication by young people will ensure that Seychelles can operate confidently within a commercial and cultural environment that increasingly relies on the rapid transfer of information by electronic means. (*EDUCATION FOR A LEARNING SOCIETY: Policy statement of the ministry of education, Seychelles* **P. *5***)
- It should provide the widest possible access to ICT by student and harnessing its full potential for supporting and enhancing the learning process, and increasing the effectiveness of management in education. (*EDUCATION FOR A LEARNING SOCIETY: Policy statement of the ministry of education, Seychelles* **P. 6***)*
- It should equip all institutions with ICT equipment and programmes for the teaching of ICT, and build capacity to eliminate ICT illiteracy in the country. *(Ministry of Education: Education Sector Development Plan 2007-2011, focus area 06: 6.1.1.f and 6.1.1.i)*
- ICT is an object of the learning process, as well as a productivity tool. *(National curriculum Framework, 2001.* **P.13***)*

- ICT teaching should be developed at all levels of the formal educational system. (*National ICT Policy for Seychelles* **P. 7***)*
- Opportunities should exist which will enable all learners to acquire ICT skills and be able to use them confidently and creatively to access employment or further training. (*National ICT Policy for Seychelles* **P. 8***)*

STRUCTURE AND FEATURES OF THE FRAMEWORK DOCUMENT

This ICT Curriculum Framework prescribes what students should know, appreciate and be able to do as they are exposed with computers in the secondary schools from S1 to S3. The main part and perhaps the most valuable section is a collection of competency statements which forms a foundation for content and practice.

Its essential purpose is to set out a framework around which schools can build teaching and learning programmes based on the agreed competencies. The framework document runs alongside the Unit Framework Plans, giving a clear structure to each unit across all the relevant secondary levels. The Unit Framework Plans are guided by clear competency statements for each application that also identifies progression through the units.

The framework consists of the following components:

- Ten competency statements
- Elements of competency for each competency statements (Outcomes and standards)
- Performance criteria for each element of competency

This competency framework document describes the range of knowledge, skills and attitudes in ICT in order for the learner to cope effectively in this current information processing age. This framework was developed through an extensive process that included widespread consultation with employers, ICT teachers, a comprehensive review of literature relating to competency based education, a thorough analysis of similar national and international competency frameworks, and last but not least a close consultation of the IGCSE ICT curriculum for 2008.

Selected competencies from the framework document can be used to provide consistent language for:

- developing other ICT programmes at school level .
- describing competency requirements for a particular student's level.
- professional development of teachers in the area of ICT literacy at school level.

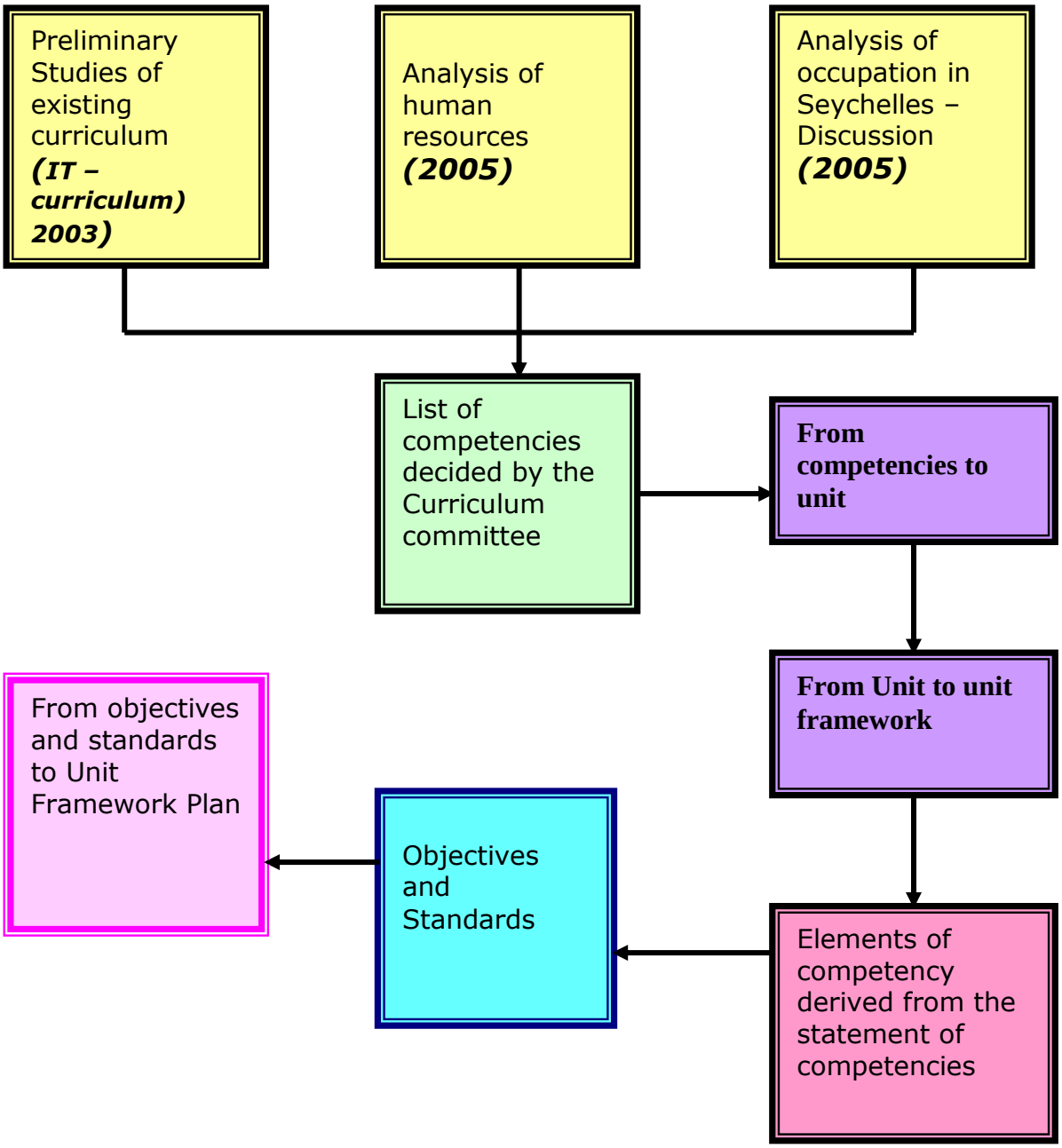
COMMON PRACTICES

The following common practices have been taken into great consideration when developing this framework document:

- The relevant stakeholders fully participated in identifying, defining, and reaching a consensus about important ICT statements of competencies.
- Statements and elements of competencies are clearly defined, understood, and accepted by relevant stakeholders.
- Elements of competencies are defined at a sufficient level of specificity that they can be assessed.
- Assessments of competencies provide useful and meaningful information that is relevant to decision-making.
- The assessments of competencies are directly linked with the goals of the learning experience.

Also the matter of "data ramifications" that ensure that competencies are both valid and reliable with the ultimate aim, in practice, being that they are fully transferable between and outside of schools has been taken into consideration when developing the competencies.

THE CURRICULUM DESIGN PROCESS



COMPETENCY BASED APPROACH

Competency-Based Education (CBE) is an approach to instruction and assessment that places primary emphasis on identifying and measuring specific learning outcomes, or competencies. A Competency Based Approach (CBA) has been used to develop this ICT curriculum framework. This is a major shift in the pedagogical philosophy, towards identification of competencies, rather than behavioral learning objectives, that do not equal students' workforce functionality. CBE is indeed an institutional process that moves education from focusing on what academics believe graduates need to know (teacher-focused) to what students need to know and be able to do in varying and complex situations (student and/or workplace focused). CBE is focused on outcomes (competencies) that are linked to workforce needs, as defined by employers and the profession. Unlike general goals, competencies are written as real-life abilities that are required for effective professional practice.

The framework is designed to allow the students to develop a clear understanding of what they are learning and the ability to understand the processes that they are going through, this includes an awareness of areas for assessment.
This approach endeavours to enhance the relevance of teaching and promote the transfer of knowledge and skills to the real life situation which could be across the curricula or the world of work. The framework also allows for clear management of the subject of ICT by the coordinator. Planning can be monitored for progression and also clear expectations can be developed within the school. By following a framework, the student's learning can be monitored with clear reference to the unit framework plans.

CBE deviates from the traditional pre-programmed transfer of knowledge, skills and attitudes. Instead it focuses on the integral development of competencies in a flexible learning process. Learners define the required knowledge, skills and attitudes to become successful in the envisaged field of work. Teachers design and facilitate flexible learning processes to provide the right opportunities for students to develop these competencies. Students continuously assess their own competencies and plan their individual learning trails to achieve full competency, while being coached and guided by their teachers.

Competency is the combination of skills, attitudes and knowledge needed to perform a specific task. Skills, abilities and knowledge are developed through learning experiences. Different combinations of skills and knowledge that one has acquired define the competencies that an individual possess. Knowledge gives the student the ability to understand things.

CURRICULUM FRAMEWORK FOCAL POINTS

The focus of the new ICT curriculum is now learner centered.
The key purpose of the new ICT curriculum can be further defined as to:

- provide a practical and realistic approach to teaching with technology.
- focus on developing positive attitude in students.
- enrich learning environment using technologies.
- integrate theoretical and practical elements in learning.
- define clear learning and teaching guidelines for students and teachers.
- encourage independent learning and self responsibility for learning.
- foster lifelong learning

With the new ICT curriculum and Competency based approach to teaching and learning, each student will be given computer to learn independently. Teacher as a facilitator diagnose student abilities to facilitate students growth and development. Teachers prepare students to perform skills they will need in ICT.

Teacher will have to act as facilitator. Each student should be given enough time to work independently. Teachers can check the practical activity done by students using check list. More practical activity should be given to get better understanding of the subject. Teachers need to prepare many practical activities. Planning is very important to make use of time very effectively. Teachers will have to plan the activities well in advance. Check list for practical is necessary to monitor the students' progress in the subject.

Non-contact hours work is an integral part of the curriculum, which provides continuity to learning and creates, stimulate and facilitate student centered learning.

FEATURES OF ICT

Importance of ICT

ICT has increasing importance within the school curriculum. Not only does it support teaching and learning within other curriculum subjects, but it is also a discipline of study in its own right. Developing skills, knowledge and understanding with the use of ICT prepares pupils to use such technologies in their everyday and working lives. ICT enable pupils to access, share, analyse and present information gained from a variety of sources and in many different ways. The use of ICT provides opportunities for pupils to work both collaboratively and independently, to consider which ICT tools best suit the task in hand and to know when and when not to use ICT to complete such a task. As such, the role of ICT as a discipline is not only to enhance the learning experiences of pupils but also to help them develop the skills essential to participate effectively, both now and in the future.

Features of ICT in the framework document

This curriculum framework document presents ICT as three interrelated areas of knowledge, skills and attitudes, which will prepare student for, cross curricula application of ICT, IGCSE examination, the world of work. The Interrelated areas are *Contemporary Skills*, *Foundational Concepts* and *Intellectual Capabilities*. It comprised of the following applications:

- Manipulating text and numerical data (word processing, spreadsheets).
- Manipulating graphic information (scanning and drawing software).
- Storing and analysing digitized information (databases).
- Accessing and disseminating information (world wide web, CD-ROMs).
- Communicating (e-mail).
- Developing Websites (Web Design).
- Presenting Information (Presentation Graphics).

Contemporary Skills (*Ability to use contemporary applications*)

Word processing

Word processing as an application program and writing tool features the processing of electronic documents in a real life situation. Student will use real documents, embedded in authentic activities, to exhibit their fluency in using a word-processing program in real life setting and situation.

Spreadsheet

Spreadsheet as an application program and cognitive tool features the inputting, calculating and presenting of data in a model. Student will use real data, embedded in authentic activities, to exhibit their fluency in using a Spreadsheet program in real life setting and situation.

Database
Database as an application program and cognitive tool, features the inputting, manipulating, retrieving and the summarizing of data in a model for real life purpose. Student will use real data, embedded in authentic activities, to exhibit their fluency in using a database program in real life setting and situation.

Computer Graphics
Computer Graphics program allows a user to develop and edit graphical interfaces that are heavily present in the real world of graphics. Student will use the program to build and edit simple graphics.

Presentation

Presentation software is a type of application program, which allows information to be presented to an audience in an effective way. Student will use application software to prepare the presentation of information.

Web Design Program

Web page design programs features the ability to produce web pages for internet communication and information access. Student will use the program to build simple web pages for Intranet or Internet use.

Telecommunications
Internet is the International network of computers sharing information. Email is the electronic posting of document from one computer to another. Students will use them to find and process information as well as for the electronic dispatch of documents.

Foundational Concepts (*The principles underpinning ICT)*

- Computer system

Intellectual Capabilities (*Application of ICT*)

- Finding information
- Planning and selecting information
- Developing information
- Presenting information
- Reviewing the use of ICT
- Problem solving

STATEMENT OF COMPETENCIES

Competency oriented curriculum is a variant of the outcome based curriculum type. The concept of competency focuses on the desired results of the student rather than on the learning process and embodies the ability to transfer and apply skills and knowledge to new situations and environments. The competencies are taken as the final goals to be acquired by the students. The final outcomes are derived from authentic tasks in real life professions.

Through the competencies listed, the framework document intend to provide a frame of reference for the outcomes with regard to knowledge, skills and attitudes that can be achieved at the different secondary levels, and thus being ICT literate, as well as responding adequately to the expectations of society and continuing education with regard to ICT competencies.

List of Competencies

1	Demonstrate understanding of basic Computer Systems - Hardware and communication.
2	Use a Word Processor to produce documents electronically.
3	Use Presentation program to present information.
4	Demonstrate an understanding of basic Computer systems – Software.
5	Use Spreadsheet program to input, manipulate and summarise data in a model.
6	Use graphics program to produce simple graphics.
7	Demonstrate understanding of basic computer systems and Network.
8	Use a Website developing program to build Website.
9	Use database program to input, manipulate, retrieve and summarise data in a model.
10	Demonstrate understanding of the nature and social impact of computer technology.

Statement of competency	Strategies
Demonstrate understanding of basic Computer System –Hardware and Communication.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion. ***With the help of*** Textbooks, Copies of relevant standard/Quality document, Flash cards ***From*** Demonstrations, research, discussions, oral presentations, Practical exercises, Software Presentation. ***Others*** Individual, Pairs. Group
Elements of competency	**Performance criteria**
Element1 1.0 Demonstrate basic knowledge of the different types of computers	1.1. Identify correctly the different types of computers. 1.2 State correctly the use of the different types of computers.

Element 2 2.0 Demonstrate knowledge of hardware and software.	2.1 Differentiate between hardware and software. 2.2 Differentiate between the different types of software. 2.3 Identify correctly the hardware. Components of the computer system. 2.4 Identify correctly the different peripheral devices. 2.5 Use appropriate name for all hardware.

Element 3 3.0 Demonstrate basic skills in operating the computer.	3.1 Start and shut down a computer properly. 3.2 Manage correctly a current Operating System. 3.3 Open and close a program within current operating system.
Element 4 4.0 Demonstrate basic understanding of the Network systems.	4.1 Differentiate between stand alone and network computers. 4.2 Demonstrate ability to access the document on a client-server in a network system. 4.3 Demonstrate the ability to access shared documents on network systems.
Element 5 5.0 Demonstrate basic understanding of World Wide Web and email.	5.1 Connect to the Internet. 5.2 Use search engines appropriately to locate learning materials and special posted information. 5.3 Extract relevant information from located learning materials and special posted information. 5.4 Create an email account. 5.5 Send emails correctly. 5.6 Retrieve emails correctly.

Statement of competency	Strategies
Statement of competency Use a Word Processor to produce documents.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion ***With the help of*** Textbooks, Copies of relevant standard/Quality documents, ***From*** Demonstrations, research, discussions, oral presentations, Practical exercises, Software Presentation. ***Others*** Individual, Pairs. Group
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate understanding of word processing concepts.	1.1 State correctly the advantages of a computer as a word processing device over a conventional typewriter. 1.2 Identify correctly different Word-Processing applications available.
Element 2 2.0 Demonstrate the ability to create documents.	2.1 Identify correctly the different screen elements**.** 2.2 Create documents using the option on the toolbar. 2.3 Enter text correctly.
Element 3 3.0 Demonstrate the ability to save and open a document.	3.1 Save a document correctly. 3.2 Open a document correctly.
Element 4 4.0 Display the ability to edit a document	4.1 use scroll bars and arrow keys to move around a document. 4.2 Select text appropriately in a

	document. 4.3 Move text using cut and paste. 4.4 Copy text using copy and paste. 4.5 use undo command appropriately to recover incorrect editing. 4.6 Employ correctly word processing spellchecking tools. 4.7 Insert special symbols in a document.
Element 5 5.0 Demonstrate the ability to create and modify a table.	5. 1 Insert a table in a document. 5. 2 Add text into a table. 5. 3 Insert and remove rows and columns in a table. 5. 4 Merge and split cells within a table.
Element 6 6.0 Demonstrate the ability to format a document.	6.1 Format characters. 6.2 Format paragraphs. 6.3 Format Page.
Element 7 7.0 Demonstrate the ability to format graphics.	7.1 Insert pictures in a document. 7.2 Insert WordArt in a document. 7.3 Format graphics in a document.
Element 8 8.0 Demonstrate proficiency in basic keyboarding skills**.**	Use appropriate keys on the keyboard.
Element 9 9.0 Demonstrate the ability to print a document.	9.1 Set up print options. 9.2 Print a document.

Statement of competency	Context of achievement ***By means of*** Exposures, Demonstration, Group discussion ***With the help of*** Textbooks, Copies of relevant standard/Quality documents. ***From*** Demonstrations, research, discussions, oral presentations, practical exercises, Software Presentation. ***Others*** Individual, group, pairs.
Use graphics program to produce simple graphics.	
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate knowledge of graphics and Computer graphics.	1.1 Differentiate between graphics and computer graphics. 1.2 State examples of application programs used for computer graphics. 1.3 State the advantages of using graphics program. 1.4 Identify appropriate hardware that support graphics software.
Element 2 2.0 Demonstrate understanding of a graphic design.	2.1 Identify the screen elements in a graphics program. 2.2 Design a graphic. 2.3 Edit the graphics by selecting the appropriate tools. 2.4 Edit and save different types of graphics format.
Element 3 3.0 Demonstrate the ability to manipulate graphics.	3.1 Import objects from other applications. 3.2 Export objects to other applications. 3.3 Rotate text and graphics. 3.4 Apply 2-D and 3-D graphics principles. 3.5 Manipulate graphical tools.

Element 4 4.0 Demonstrate understanding of typography for graphics design.	4.1 Select typefaces. 4.2 Use unit conversions correctly. 4.3 Select fonts and line spacing. 4.4 Select type formats.

Statement of competency	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion ***With the help of*** Textbooks, Copies of relevant standard/Quality documents. ***From*** Demonstrations, research, discussions, oral presentations, practical exercises, Software Presentation. ***Others*** Individual, group, pairs.
Demonstrate understanding of basic computer systems and software.	
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate understanding of peripheral devices.	1.1 Identify peripheral devices. 1.2 State the use of peripheral devices.
Element 2 2.0 Demonstrate the ability to use storage devices.	2.1 Locate the different storage devices. 2.2 Use different storage devices correctly (Pen-drive, CD ROM, Floppy disk). 2.3 Format storage devices. 2.4 Identify the storage capacity of disks.

Element 3 3.0 Demonstrate knowledge of software.	3.1 Identify available software. 3.2 State the functions of available application software. 3.3 Demonstrate awareness of software copyright.
Element 4 4.0 Demonstrate understanding of operating systems.	4.1 Demonstrate knowledge of the two main operating systems (PC and Network). 4.2 Differentiate between Character User Interface and Graphical User Interface.
Element 5 5.0 Demonstrate understanding of effective file management.	5.1 Demonstrate understanding of files concepts. 5.2 Demonstrate understanding of folders concept. 5.3 Locate files on storage media. 5.4 Copy, move and delete files. 5.5 Create shortcuts to files and folders.

Statement of competency	Strategies
Statement of competency Use Spreadsheet program to input, manipulate and summarise data in a model.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion ***With the help of*** Textbooks, Copies of relevant standard/Quality documents. ***From*** Demonstrations, research, discussions, oral presentations, Practical exercises, Software Presentation. ***Others*** Individual, Group, Pairs.

Elements of competency	Performance criteria
Element 1 1.0 Demonstrate understanding of spreadsheet concepts.	1.1 State advantages of electronic spreadsheet over manual spreadsheet. 1.2 Explain the uses of spreadsheet in real life. 1.3 Plan a spreadsheet appropriately.
Element 2 2.0 Use a spreadsheet program to enter different kinds of data.	2.1 Show understanding of cell references. 2.2 Enter model data in a worksheet. 2.3 Identify cell content. 2.4 Use number format to display data.
Element 3 3.0 Modify cell contents.	3.1 Edit cell contents. 3.2 Demonstrate understanding of cell references. 3.3 Copy and move cell contents.
Element 4 4.0 Use basic spreadsheet formulae and functions.	4.1 Use basic mathematical operators to construct simple formulae. 4.3 Use basic functions.
Element 5 5.0 Use graph to present data.	5.1 Select chart types to present data. 5.2 Format graphs.
Element 6 6.0 Format a worksheet.	6.1 Format a row. 6.2 Format a column. 6.3 Format worksheet appropriately.
Element 7 7.0 Print a worksheet	7.1 Print preview a worksheet 7.2 Print a specific area of the worksheet.

Statement of competency Use Presentation program to present information.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion ***With the help of*** Textbooks, Copies of relevant standard/Quality documents, ***From*** Demonstrations, research, discussions, oral presentations, practical exercises, Software Presentation. ***Others*** Individual, Group, Pairs.
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate understanding of Presentation software and hardware.	1.1 State advantages of electronic presentations. 1.2 Identify the use of Presentation software in real life situation. 1.3 Identify hardware items that support presentation software.
Element 2 2.0 Demonstrate understanding of presentation process.	2.1 Use the four-steps of processes in preparing a presentation. 2.2 Identify different ways to present information.
Element 3 3.0 Demonstrate the ability to design a presentation.	3.1 Create a presentation. 3.2 Add text, graphics and tables in a slide. 3.3 Format text, graphics and tables in a slide.

Element 4 4.0 Demonstrate the ability to arrange and control the presentation.	4.1 Organise slides. 4.2 Animate objects in a slide. 4.3 Test animation and transition effect in a slide.
Element 5 5.0 Demonstrate the ability to run a presentation.	5.1 Set up slides for a presentation. 5.2 Run a slide show.
Element 6 6.0 Use print options to print a presentation.	6.1 Print preview a presentation. 6.2 Print relevant presentation information.

Statement of competency	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion ***With the help of*** Textbooks, Copies of relevant standard/Quality documents. ***From*** Demonstrations, research, discussions, oral presentations, practical exercises, Software Presentation. ***Others*** Individual, Group, Pairs.
Demonstrate understanding of basic computer systems and Network.	
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate knowledge of functions of Central Processing Unit (CPU) and memory.	1.1 State the importance of CPU. 1.2 State the importance of memory. 1.3 Differentiate between Read only Memory (ROM) and Random Access Memory (RAM).

Element 2 2.0 Demonstrate understanding of resolution.	2.1 Change Screen resolutions appropriately. 2.2 Change printer resolutions appropriately.
Element 3 3.0 Demonstrate knowledge of the uses of Liquid Crystal Display (LCD) Screens.	3.1 State uses of Liquid Crystal Display screen. 3.2 State advantages of using Liquid Crystal Display Screens.
Element 4 4.0 Demonstrate knowledge of network systems.	4.1 Demonstrate understanding of different types of network system and its advantages. 4.2 Demonstrate understanding of the advantages and disadvantages of Local Area Networks (LAN). 4.3 Demonstrate understanding of the advantages and disadvantages of Wide Area Networks (WAN). 4.4 Differentiate Local Area Networks from Wide Area Networks.
Element 5 5.0 Demonstrate understanding of the impact of using ICT.	5.1 Define computer virus. 5.2 State correct measurements that must be taken in order to protect against hacking and viruses. 5.3 Demonstrate understanding of the potential health problem related to the prolonged use of ICT. 5.4 Demonstrate awareness of business and society change as a result of computer development. 5.5 Demonstrate awareness of the reliance of society on ICT.
Element 6 6.0 Demonstrate understanding of different programming languages.	6.1 Use programming languages in context. 6.2 Differentiate the functions of interpreter and compiler.

Element 7 7.0 Demonstrate knowledge of the different applications of ICT.	7.1 Identify uses of ICT in Medical administration. 7.2 Identify uses of ICT in School administration. 7.3 Identify uses of ICT in Shops and Supermarkets.

Statement of competency	Strategies
Statement of competency Use a Website developing program to build Websites.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion. ***With the help of*** Textbooks, Copies of relevant standard/Quality documents, ***From*** Demonstrations, research, discussions, oral presentations, Practical exercises, Software Presentation. ***Others*** Individual, Group, Pairs.
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate knowledge of basic web design.	1.1 Identify the fundamentals of Webpage. 1.2. Plan a Webpage.
Element 2 2.0 Demonstrate understanding of website and web page.	2.1 Differentiate different types of Websites. 2.2 Differentiate between webpage and website.
Element 3 3.0 Demonstrate the ability to create a website.	3.1 Create a webpage. 3.2 Hyperlink web pages. 3.3 Create website. 3.4 Test the hyperlinks.

Statement of competency	Strategies
Statement of competency Use database program to input, manipulate, retrieve and summarise data in a model.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion. ***With the help of*** Textbooks, Copies of relevant standard/Quality documents. ***From*** Demonstrations, research, discussions, oral presentations, Practical exercises, Software Presentation. ***Others*** Individual, Group, pairs.
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate basic understanding of an electronic database.	1.1 Define a database. 1.2 Identify correctly instances where it is appropriate to use a database. 1.3 Identify the advantages of using a database.
Element 2 2.0 Demonstrate understanding of a table object.	2.1 Identify the structure of a table 2.2 Plan the necessary fields in a table 2.3 Create a table.
Element 3 3. 0 Demonstrate understanding of simple entry forms.	3.1 Create a simple form. 3.2 Use forms to add records to a table. 3.3 Modify forms. 3.4 Use form controls.
Element 4 4.0 Demonstrate ability to create simple select queries.	4.1 Select fields for query. 4.2 Apply criteria for query. 4.3 Run a query.
Element 5 5.0 Demonstrate the ability to produce simple reports.	5.0 Describe the importance of a report. 5.1 Create a simple report. 5.2 Modify a report. 5.3 Print report.

Statement of competency	Strategies
Demonstrate understanding of the adverse impact of ICT on society.	**Context of achievement** ***By means of*** Exposures, Demonstration, Group discussion. ***With the help of*** Textbooks, Copies of relevant standard/Quality document, Flash cards ***From*** Demonstrations, research, discussions, oral presentations, Practical exercises, Software Presentation. ***Others*** Individual, Pairs. Group
Elements of competency	**Performance criteria**
Element 1 1.0 Demonstrate understanding of computer malpractice and crime.	1.1 State illegal uses of an ICT system (e.g., Pen drives, Mobile phones, CDs, Internet, etc..) 1.2 Demonstrate knowledge of professional misconduct with a computer system. 1.3 Demonstrate knowledge of Hackers and Crackers. 1.4 Demonstrate knowledge of computer crimes committed by Hackers and Crackers. 1.5 Demonstrate knowledge of damage caused by computer viruses. 1.6 State measure that organization can take to protect their system.

Element 2 2.0 Demonstrate understanding of legal and ethical issues.	2.1 Demonstrate knowledge of MOE's policy/guidelines pertaining to ICT use. 2.2 Demonstrate awareness of national policy/guidelines pertaining to ICT use. 2.3 Demonstrate knowledge of the potential for malpractice and crime using the Internet. 2.4 Demonstrate knowledge that internet access can put children at risk. 2.5 Demonstrate knowledge that fraud can be made by using websites (e.g, bogus and auction sites).

CONTACT HOURS

Statement of competency	**Unit title**	**Contact hours**		**Term (s) involved**
		Weekly	**Termly**	
Demonstrate understanding of basic Computer Systems, Hardware, and communication.	Computer System -	2 hrs 40 mins (4 periods 32 hrs))	**28 hrs (About 5hrs per element**)	1
Use a Word Processor to produce documents electronically.	Word processing	2 hrs 40 mins	45 hrs	1,2 and 3 **6hrs T1 32hrs T2 7hrs T3**
Use graphics program to produce simple graphics	Computer graphics	2 hrs 40 mins	23 hrs	3
Demonstrate an understanding of basic computer systems and Software.	Computer System Level 2	2 hrs 40 mins	28 hrs	1
Use Spreadsheet program to input, manipulate and summarise data in a model.	Spreadsheet	2 hrs 40 mins	40hrs	1,2 and 3
Use Presentation program to present information.	Presentation	2 hrs 40 mins	28 hrs	3
Demonstrate understanding of basic computer systems and network.	Computer System Level 3	2 hrs 40 mins	44 hrs	1,2

Statement of competency	Unit title	Contact hours		Term (s) involved
		Weekly	Termly	
Use a Website developing program to build website	Web Design	2 hrs 40 mins	26 hrs	2
Use database program to input, manipulate, retrieve and summarise data in a model.	Database	2 hrs 40 mins	26 hrs	3

Unit Framework Plan

The unit framework gives the big picture for the development of the various teaching units. It describes the Unit code, Unit title, Statement of competency for the respective unit, level and its teaching term.

Unit code	Unit Title	Statement of competency	Level	Term
ICTCS01	Computer System Level 1	Demonstrate an understanding of basic Computer Systems, hardware and communication.	S1	1
ICTWP01	Word processing	Use a Word Processor to produce documents electronically .	S1	2
ICTGP01	Computer graphics	Use Graphics program to produce simple graphics.	S1	3
ICTCS02	Computer System Level 2	Demonstrate an understanding of basic computer system and software.	S2	1
ICTSP01	Spreadsheet	Use Spreadsheet program to input, manipulate and summarise data in a model.	S2	2
ICTPG01	Presentation	Use Presentation Graphics program to	S2	3

		produce a slide show.		

Unit code	Unit Title	Statement of competency	Level	Term
ICTCS03	Computer System Level 3	Demonstrate an understanding of basic computer system and network.	S3	1
ICTWD01	Web Design	Use a website developing program to build web pages.	S3	2
ICTDB01	Database	Use database program to input, manipulate, retrieve and summarise data in a model.	S3	3

TERMLY BREAK DOWN

Secondary one (S1)		
Term 1	**Term 2**	**Term 3**
Demonstrate understanding of basic Computer System and communication. (28 hrs)	Use a Word Processor to produce documents electronically. (45 hrs)	Use Presentation program to present information (23 hrs)

Secondary two (S2)		
Term 1	**Term 2**	**Term 3**
Display basic knowledge of peripheral devices and their uses. (28 hrs)	Use Spreadsheet program to input, manipulate and summarise data in a model (40hrs)	Use graphics program to produce simple graphics (28 hrs)

Secondary three (S3)		
Term 1	**Term 2**	**Term 3**
Demonstrate understanding of basic computer systems and network. (44 hrs)	Use a Website developing program to build Website. (26 hrs)	Use database program to input, manipulate, retrieve and summarise data in a model (26 hrs)

Number of contact hours/non contact hours per year		
S1-Year 1 :96/24 Termly: 32/ 8	S2- Year 2: 96/24 Termly: 32/ 8	S3- Year 3: 96/24 Termly: 32/ 8

Collection of information for the Final Assessment Project will be done during non contact hours.

TEACHING AND LEARNING STRATEGIES

- *"In competency-based teaching, you help students learn how to learn, rather than just cover content. It is the responsibility of the teacher to assist the students in acquiring the knowledge or skill before they leave the instructional experience"*

This curriculum framework approach is grounded in student-centred learning where learning is required to be learner-focused, collaborative, experiential, building on prior learning, self-managed, cumulative, and authentic. It describes an all-inclusive ICT content that will prepare young people to survive in our technological society. To achieve this end, however, all students must be given sufficient opportunities to develop the ICT competencies for each year level. What is required is a comprehensive ICT content structure of high quality and consistency throughout the grade levels. Most important, it implies quality instruction and positive, productive interactions between teachers and students. To create high expectations for quality instruction, ICT teachers must continually strive to improve their effectiveness. Competency based instruction measures what participants have learned as opposed to what instructors think they have taught.

Model Lesson

- Teachers create an optimal ICT learning environment and an effective learning experience for their students through careful planning.
- The teacher implements the lesson plan and makes adjustments as necessary for students not experiencing success or not participating.
- Instruction is developed around stated performance criteria that can be observed or measured.
- Learning is measured according to how well the learner performs in relation to competencies (objectives) rather than in relation to other learners.
- All learners have the opportunity to succeed.
- Learners receive immediate feedback on how much they have learned.
- Instruction is individualized as much as possible, with a social constructivist learning perspective in mind.

Variety of Strategies

Teachers should use a variety of teaching styles, including teacher-directed and student-initiated learning, to satisfy the learning styles of individual students and stimulate higher levels of thinking and creativity. Teachers should also employ a variety of instructional approaches, including competency-based learning, contract learning, drill and practice, information processing, lecture, mastery learning, programmed instruction, and role-playing. Especially appropriate for ICT instruction is cooperative learning. Assigning students to small learning groups (three to six members each) allows teachers and students to achieve a variety of goals, both social and cognitive.

TEACHING APPLICATIONS SOFTWARE

When it comes to learning students to use computer software, the objectives of doing so need to be considered carefully. Is the objective to teach them to become proficient or an expert in using a particular package or is the objective to allow them to see the value of the program and learn generic skills and concepts associated with it? This may determine whether the learning strategy centres around learning the command or completing tasks which develop the skills and concepts. It is clear that the students are more likely to learn these skills and concepts while applying the software to a credible task or activity. Therefore the objective of students learning software is not for them to become experts, but for them to see the value of this software and then to learn generic skills and concepts associated with them. The teachers are encouraged to use student centred activities, and approaches which is well supported by constructivist theory for the delivery of this unit.

Teaching and Learning Strategies to Meet Special Needs

Every school includes students who are high achievers; those who are low achievers; and the majority, are in the middle. The ICT teaching and learning strategies used should take into account the diverse needs of the heterogeneous groups. All students should receive feedback and reinforcement from the teacher or other students; and all should have opportunities to be involved.

Gender Equity in ICT

Students of both sexes should be treated equally throughout all the ICT classes. All students, male and female, should be given the teacher's attention and feedback on an equal basis and should be called on equally to demonstrate new skills. The teacher should also discipline the male and female students equally.

Students with Disabilities

There could be students who will attend ICT classes with motor or perceptual deficits, or other more severe disabilities. The success of those students will depend on the teacher's attitude and skill in providing instruction and support to them. The teacher should continually encourage all students to learn and experience maximum enjoyment in the ICT class by understanding students' specific needs and encouraging students who are not disabled to accept and support those who are.

COMPETENCY BASED ASSESSMENT AND STRATEGIES

Competency-based assessment is largely activity-based and practical, and competence also implies that an individual displays an understanding of the knowledge that underpins the practical performance of a task. Assessment in this competency-based programme is the process of collecting evidence and making judgements on the students' progress towards satisfying the performance criteria set out. At the appropriate point, you have to make judgment as to whether competency has been achieved.

The assessment procedures should be consistent with pedagogical philosophy adopted. The pedagogical philosophy for the delivery of ICT in this unit requires you to take a more developmental approach to learning, and a need for more formative continuous assessment. The assessing of competency in this unit should be able to accurately describe what a student can do using a Presentation program, in order to predict what they are likely to perform in the world of work, in other contexts or other circumstances. The students' developing competence (Knowledge, Skills and Attitudes) could be collected via a range of realistic procedures such as those listed below and the procedures should be relevant to the methods used to deliver the competencies in this unit. The context of assessment should ensure that evidence relating to the competency can be collected. In some cases the competency may be assessed in a simulated setting or shall be observed while task are being undertaken whether individually or in group.

Students should be assessed according to whether or not they have achieved the performance criteria for each element specified in the units of competency they are undertaking. Students are assessed either as competent, or not yet competent. It is up to the teacher to make sure that all aspects of the assessment process are, valid, reliable, fair, and flexible.

Assessment is based on the achievement of the competency described in the performance criteria. Performance criteria will measure the specific competencies. In Norm referenced assessment, very few will achieve and majority fails. Criterion referenced assessment is based on activity. Most of the students will get opportunity to score high marks and better understanding of the subjects. Students participation is more and it is students centered.

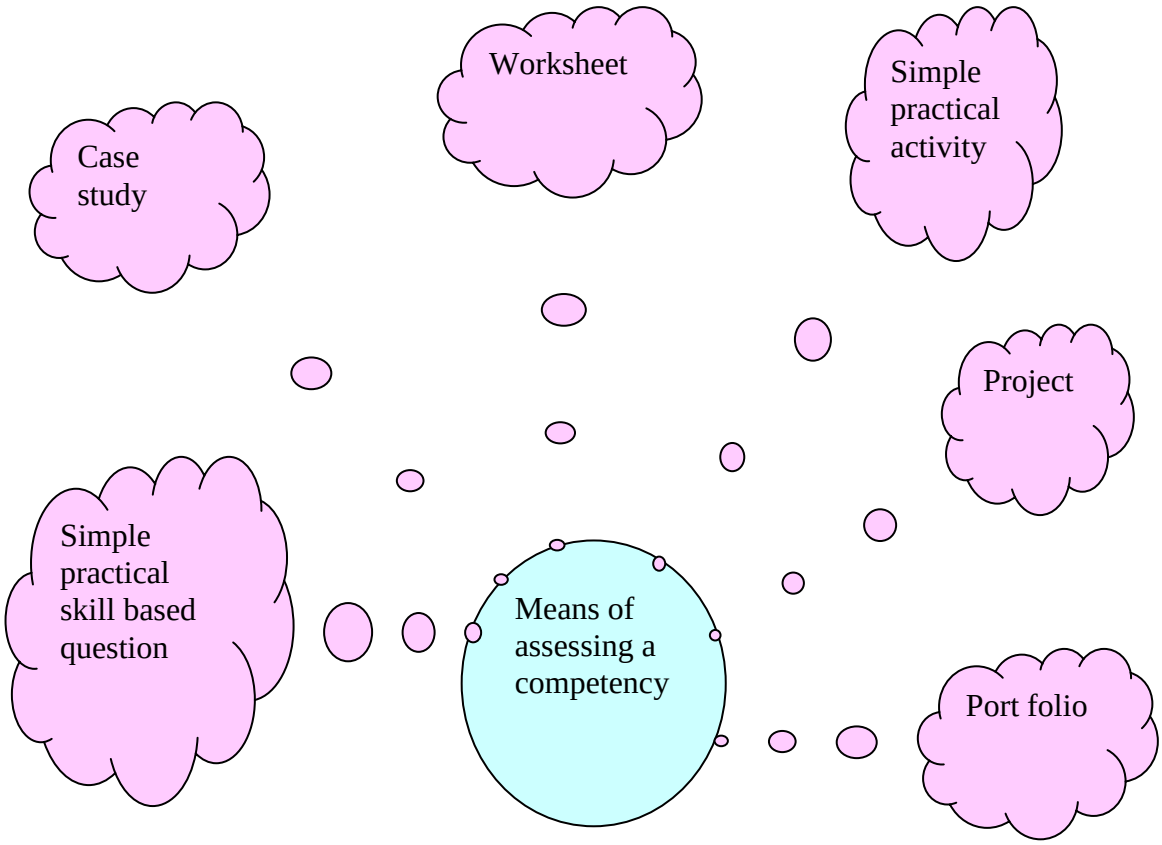
Each elements of competency will have performance criteria. Assessment will be done based on the performance criteria for that element of competency. Teachers will have to prepare a checklist for the practical skill and do practical skill tests. At the end of the unit, final assessment will have to be conducted. This final assessment should link all the elements of competency covered in that units.

INSTRUMENTS OF ASSESSMENT

Performance criteria or indicators should be known by the student. Teachers will have to select relevant performance criteria for assessment, and Write sentences describing the levels of performance. Appropriate weightings should be given based on the nature and purpose of the competency, clearly explaining the expectations of the teacher.

Several different forms of assessment can be used to measure student progress and achievement in the competencies, and they are as follows: student *portfolios, teacher observations, performance tests, open-ended questions, interviews, exhibitions, and cooperative group projects*. The results of these assessments will demonstrate how students progress through the skills, knowledge, and attitudes over time.
Self-assessment is another means of providing students with information about their performance. Students can apply self-evaluation when given lists of performance objectives and directed to judge their own achievement. This procedure can be facilitated by the videotaping of the student's performance for follow-up viewing by the student.

Student exit profile should be documented and maintained to monitor students' in the different teaching units.



ICT INSTRUCTIONAL MEDIA AND RESOURCES

Many types of instructional media and resources are available to aid the ICT teacher. In addition to textbooks, and more conventional printed materials, resources include videos, CD-Rom, and large-screen using LCD projectors. Examples abound of appropriate instructional technology for ICT. Digital camera can be used to capture students' performances for skill or strategy analysis. ICT teachers should look for ways not only to incorporate a variety of instructional media into their teaching, but also to make sure that the media are available in their teaching institutions. Instructional resources should reflect and support the vision, and guidelines described in this framework.

The following Instructional media are necessary for the delivery of the curriculum

Instructional Materials Required

Essential
Whiteboard and OHP
Demonstration computer for teacher
1:1 Student computer ratio should be maintained
Relevant updated books to be used by teachers and students
Desirable
LCD projectors

These resources are recommended for the effective delivery of the ICT curriculum.

Computer Peripherals Required

Essential
Networked computer system with Internet Access and Printers
Appropriate peripherals required to deliver the units

Computer Software Required

A wide variety of computer software is necessary for the delivery of the curriculum and it is as follows:

Essential	
Word-Processing	Database
Spreadsheet	Website Development
Presentation	Computer Graphics
Internet Browser	Email

Student Exit Profile at S3

Definition of ICT: ICT is basically the use of computer technology to create, manipulate, store, exchange and transfer data in a variety of forms (including textual, graphical, audio, video).

Areas

Computer system 1	Word processor	Presentation
Demonstrate knowledge of the different types of computers Demonstrate knowledge of hardware and software Develop basic skills in operating the computer Demonstrate basic knowledge of the Network systems	Show understanding of word processing concepts Demonstrate the ability to create documents Demonstrate the ability to save and open a document Display the ability to edit a document Demonstrate the ability to create and modify a table Demonstrate the ability to create and modify a table Demonstrate proficiency in keyboarding skills. Show the ability to print a document	Demonstrate understanding of Presentation software and hardware Show understanding of presentation process Demonstrate the ability to design a presentation Demonstrate the ability to arrange and control the presentation. Demonstrate the ability to run a presentation.

Computer system *2*	**Spreadsheet**	**Computer Graphics**
Display basic knowledge of peripheral devices and their uses Show understanding of the use of different storage devices Exhibit knowledge of software Display knowledge of operating systems Demonstrate understanding of effective file management	Show understanding of spreadsheet concepts Use a spreadsheet program to enter different kinds of data Appropriate modification of cell contents Use basic spreadsheet formulae to manipulate data Use chart/graph to present data in different forms Format a worksheet Print a worksheet	Demonstrate understanding of graphic and Computer graphic Demonstrate the ability to design a graphics Demonstrate how to manipulate graphics Demonstrate how to apply knowledge of typography

Computer system *3*	**Web design**	**Database**
Demonstrate knowledge of functions of CPU (Central Processing Unit) and memory Show understanding of resolution Show awareness of the uses of LCD (Liquid Crystal Display) Demonstrate understanding of network systems. Demonstrate the effects of using IT Demonstrate awareness of different programming languages. Show understanding	Demonstrate understanding of basic web design. Demonstrate understanding of website and web page. Demonstrate the ability to create a website Demonstrate the ability to design a web page Demonstrate basic understanding of mark up language text.	Identify instances where it is appropriate to use a database Demonstrate understanding of a table structure. Demonstrate understanding of simple entry form Demonstrate understanding of simple select queries Demonstrate the understanding of producing reports

of the importance of computers		

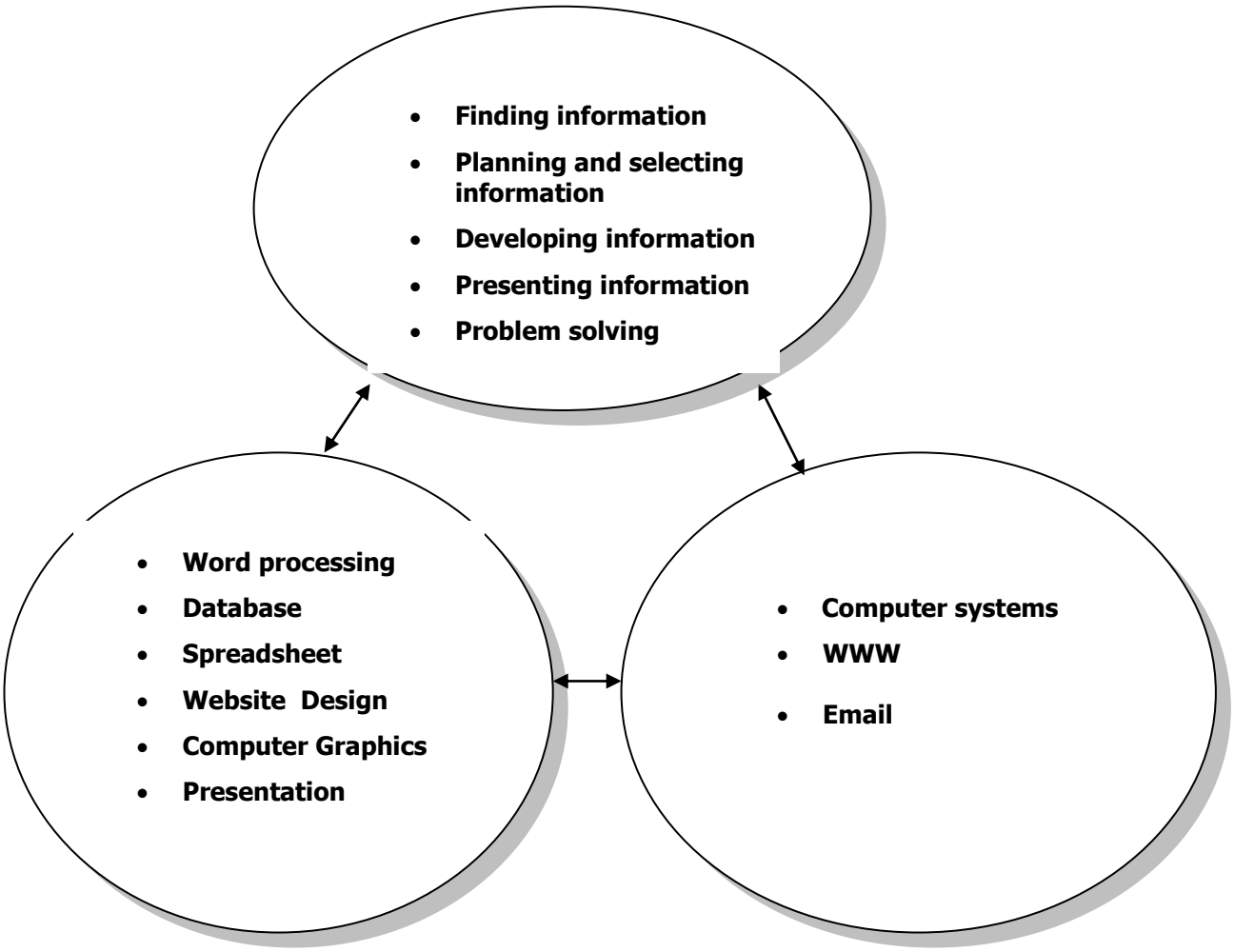
Attitudes

Responsible	Reliable	Flexible	Understanding Hardworking	Creative

Educational intentions :

- Develops thinking and problem solving skills
- Keep abreast with new technologies
- Meeting the educational needs of the student and society.

STUDENT EXIT PROFILE IN DIAGRAM



TRAINING AND INSTRUCTION FOR ICT TEACHERS

The ICT teacher should be a confident computer-user who transmits positive attitudes to students. The attitude is at least as important as the level of skill. All ICT teachers thus require computer literacy training, unless they already possess it. Teachers should be trained to help students acquire the necessary competency, and the skills common to the use of all software packages. A smaller number of specialized "ICT teachers" who are charged with teaching IGCSE ICT will require this training in greater depth.

Teachers of individual subjects should be trained in the pedagogical applications of ICT. Subject area specialists, curriculum officers, etc., would need such training in greater depth to take leading roles in matters such as in-service professional development, etc.

IMPLEMENTATION TIMETABLE AND THE FINAL STATEMENTS

The Curriculum Team determined the basic nature of the framework and presented its first draft to the NCAC in December 2006. Copies of the draft were then sent for review to some independent reviewers in 2007. After further changes were made at the request of NCAC and the independent reviewers, the framework was finalised in September 2007. The framework was presented to NCAC for approval in December 2007.

In 2008, the curriculum framework is expected to be fully implemented in all the secondary schools. This 2008 ICT framework document is endorsed by NCAC, subject to a review and revision in 2011. The framework document will be reviewed from time to time, and feedback from ICT teachers and others using this framework will always be welcome to ensure its continued relevance and to further enhance its clarity and usefulness.

Glossary

Activity	A set of tasks related either by topic, dependencies, data, common skills, or deliverables.
Assessment	A regular process that collects evidence about progress towards a goal and makes a judgment about whether this goal has been reached.
Attitude	A settled opinion or way of thinking
Competence	The ability to carry out a task or activity well enough to meet a specified standard.
Curriculum	An integrated learning programme. The curriculum describes the objectives of training, expressed in terms of learning outcomes, and how they will be assessed.
Learning outcomes	Training objectives defined as part of the curriculum. These define what the trainee will know, understand, describe, recognise, be aware of, and be able to do at the end of the training programme.
Performance	The ability to carry out a task or activity
Task	A piece of work, especially one done regularly.
Element of competency	The building blocks that make up a unit of competency. They describe groups of tasks against which people are assessed.
Performance criteria	For each element in a unit of competency, performance criteria describe the tasks against which people are assessed means the part of a competency standard specifying the required level of performance in terms of a set of outcomes which need to be achieved in order to be deemed competent